



Chicago Crime Data Analysis - 2023



3K



Total_crimes

231



Arrested_crimes

8.24



Crime Rate

672



Domestic_crimes

0.22

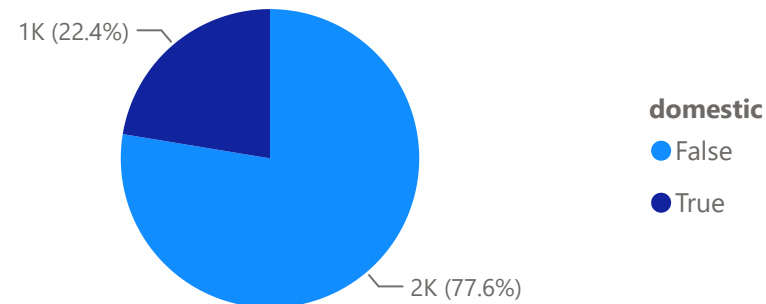


Domestic_crime_rate

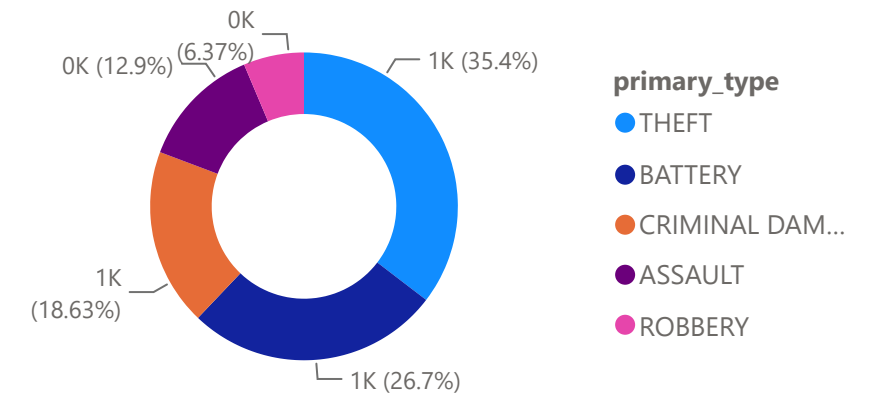
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June
- ☐ July
- ☐ August
- ☐ September

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8

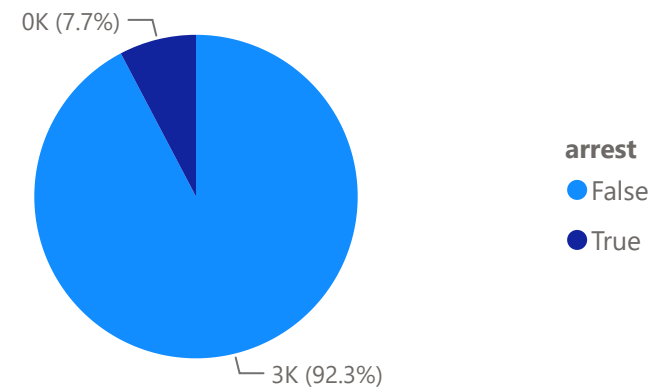
Total_crimes by domestic



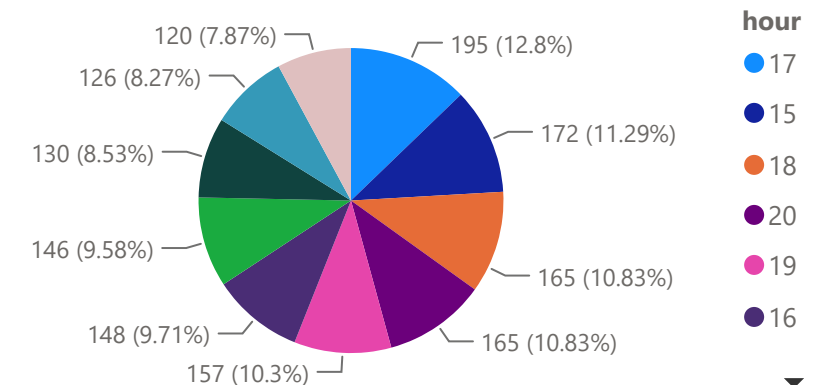
Total_crimes by primary_type



Total_crimes by arrest



Total_crimes by hour





Geographical Analysis and Hotspots

- primary_type**
- ASSAULT
 - BATTERY
 - CRIMINAL DAMAGE
 - ROBBERY
 - THEFT

76

Total_crime_locations

21

Total_Districts

50

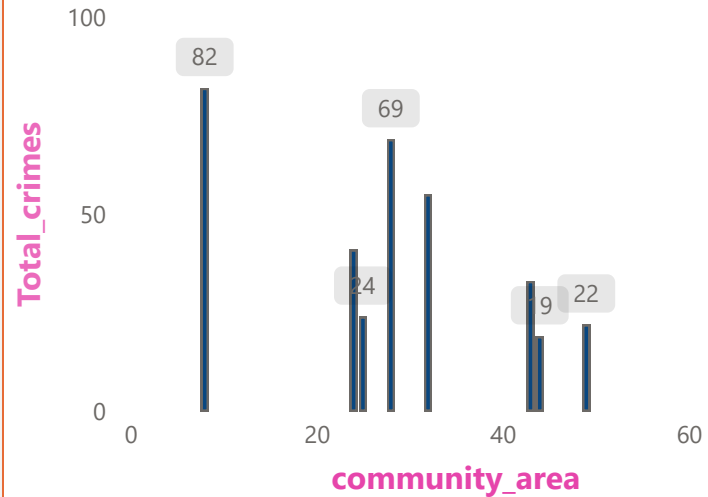
Total_Wards

74

Total_community_areas

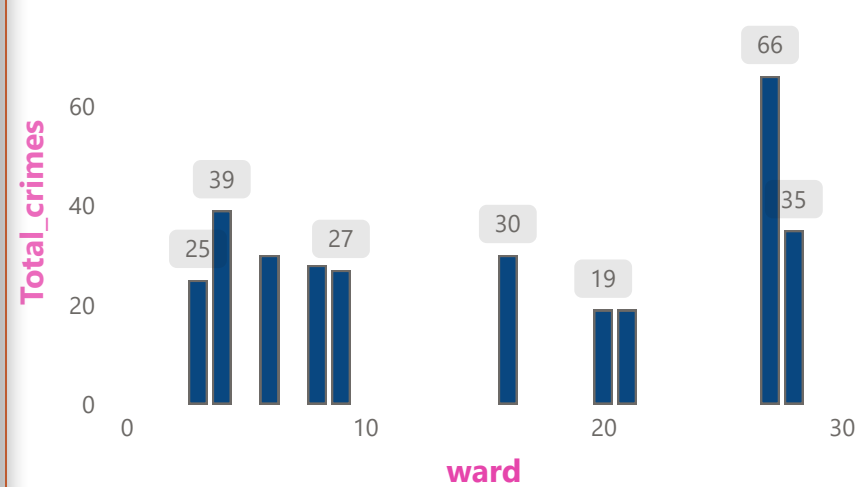
Total_crimes

by community_area



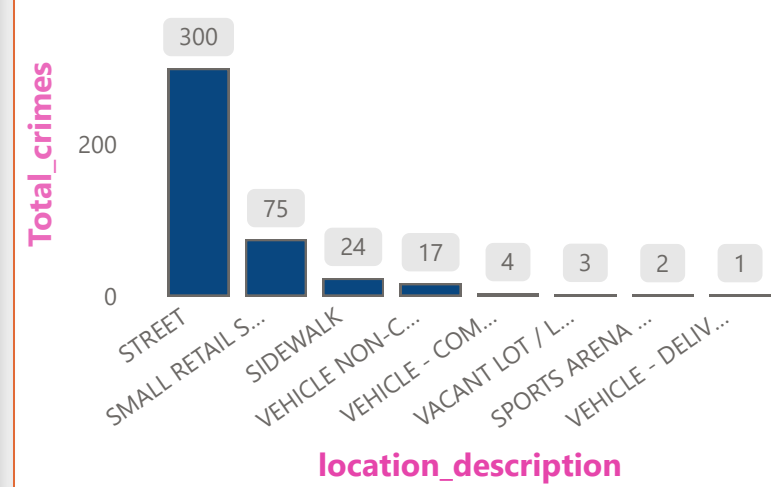
Total_crimes

by ward



Total_crimes

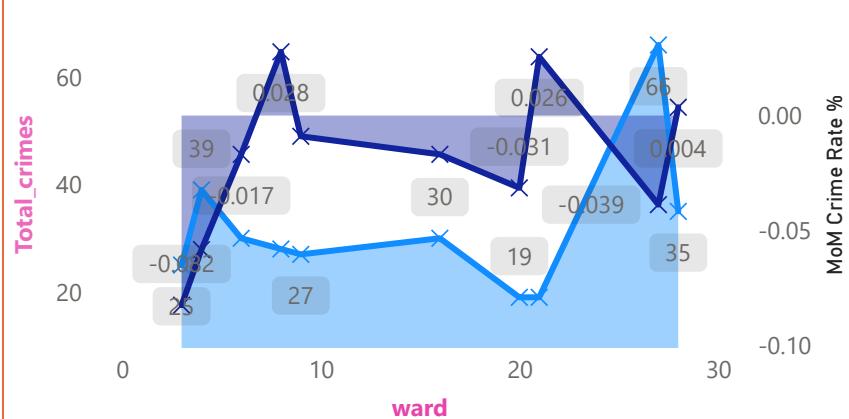
by location_description



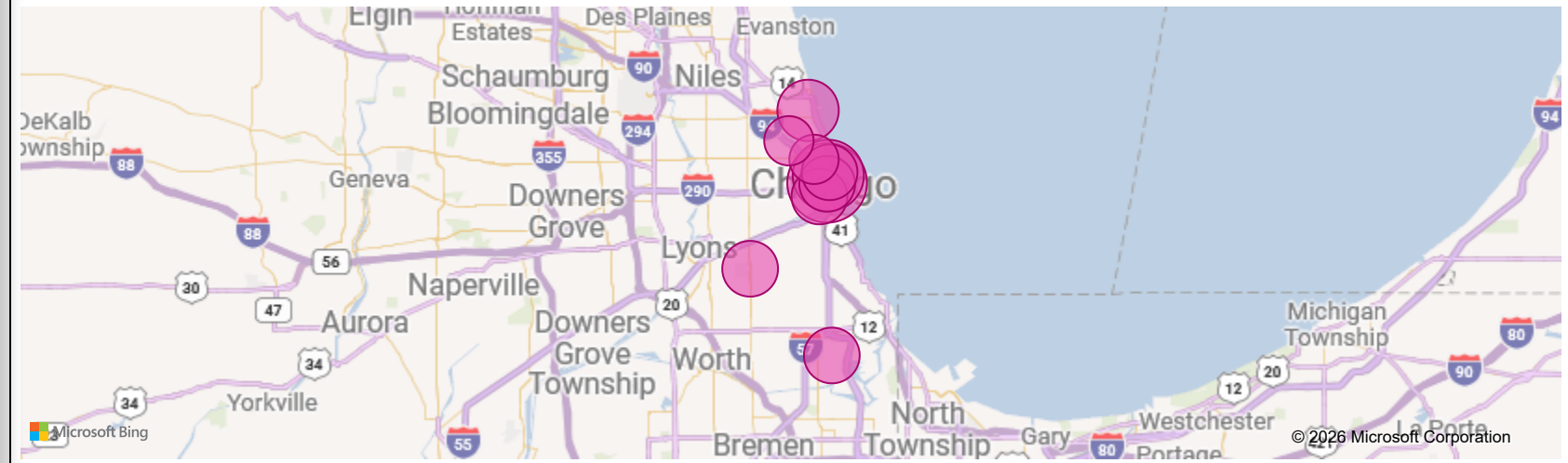
Total_crimes and MoM Crime Rate %

by ward

× Total_crimes × MoM Crime Rate %



Total_crimes by latitude and longitude





Time based Analysis

-0.01

MoM Crime Rate %

0.08

Arrest_rate

5

Primary_crimes



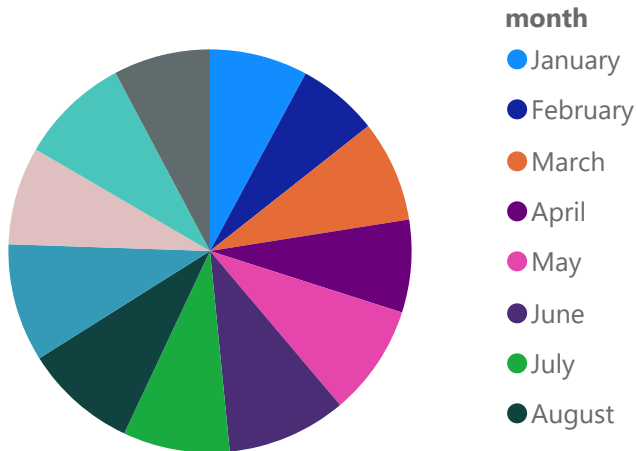
primary_type

- ☐ ASSAULT
- ☐ BATTERY
- ☐ CRIMINAL DAMAGE
- ☐ ROBBERY

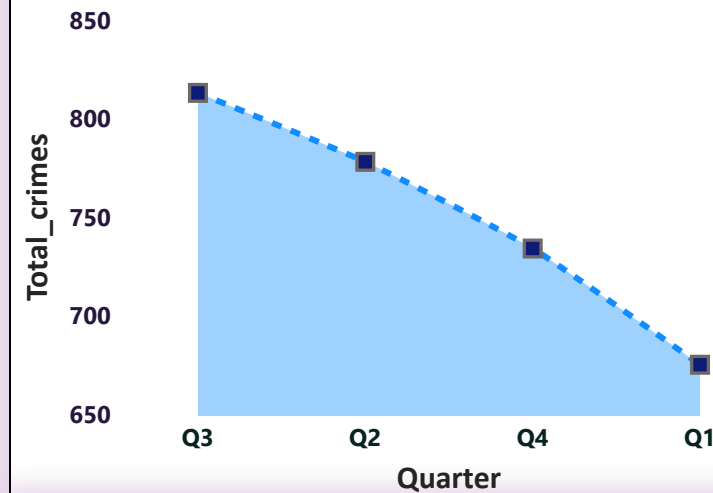
district

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

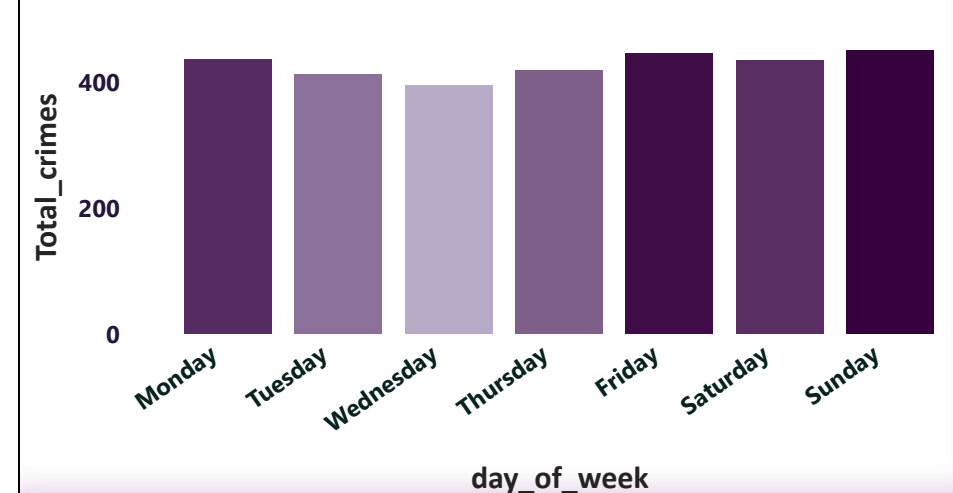
MoM Crime Rate %



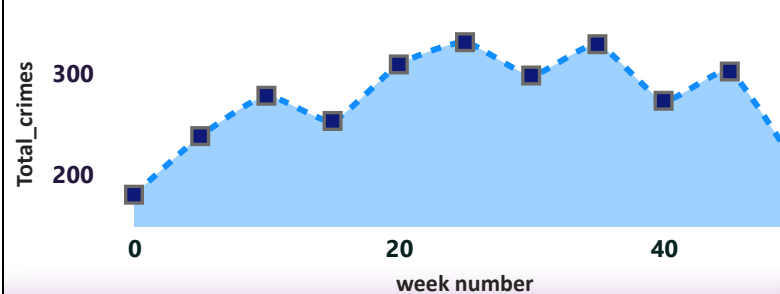
Quarter wise crimes



Day wise crimes



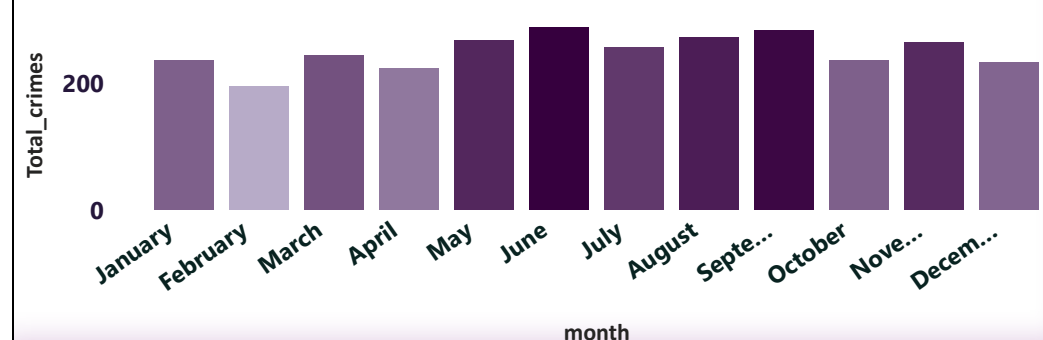
Total_crimes by Week Number (bins)



Weekend calculations

day_of_week	No	Yes	Total
Monday	437		437
Tuesday	414		414
Wednesday	395		395
Thursday	420		420
Friday	447		447
Saturday		43	436
Total	211	88	3000
	3	7	

Month wise crimes





-27K

Projected_crimes

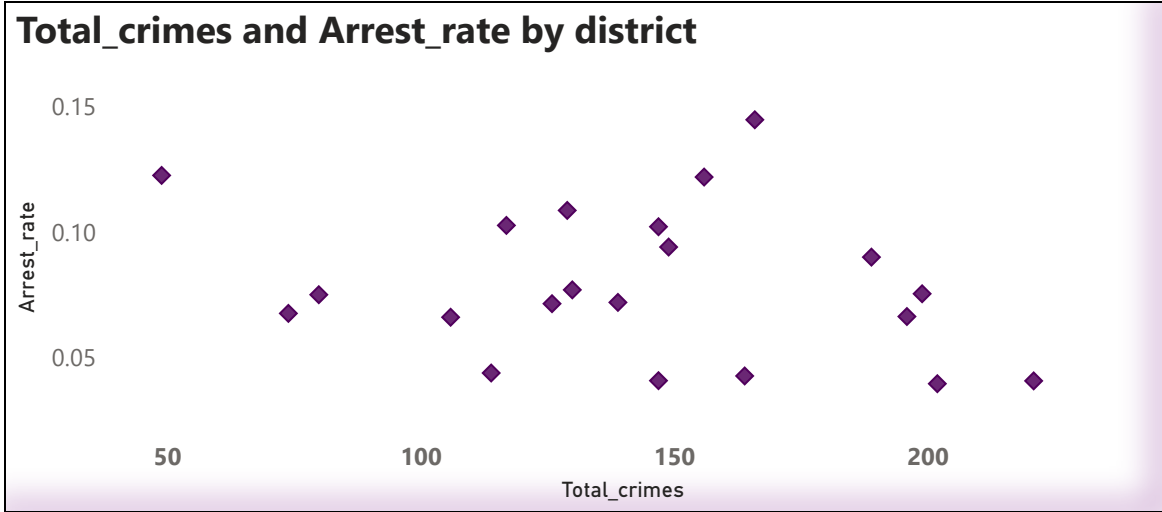
Over_view of crimes

Hotspots

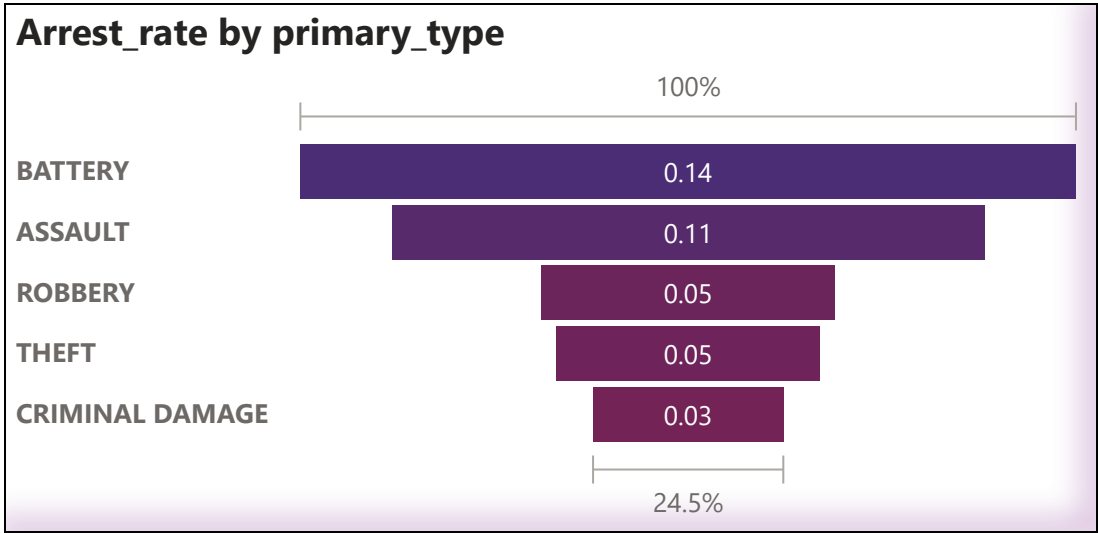
Time based Analysis

Whaf if scenario

Reduction %					
0	10	20	30	40	50
5	15	25	35	45	



district	Domestic_crimes	arrest
1	13	False
1	4	True
2	31	False
2	4	True
3	52	False
3	4	True
4	47	False
4	10	True
Total	672	



- Key findings** : only 7.7% of crimes ended in an arrest. Theft is the largest type at 35.4%. Five districts account for about 34% of crimes. 22.4% of crimes are domestic.
- What it means: crime is concentrated in a few types and a few districts, and most crimes end without an arrest.
- Recommendations: focus patrols on districts **12, 8, 6, 4 and 19**. Add resources at the peak hours and days you find. Target prevention at theft. Review why arrest rates are low for the types where they are lowest.
- The projected crimes were held on **2023** in various places in chicago city, were every crimes were investigate by **FBI**
- Crimes were recorded on all 364 days, averaging about 8 per day. Even the quietest weekday (**Wednesday, 395**) is close to the busiest (**Sunday, 451**), so coverage should stay consistent across the week and not be cut on any one day.